BIO SWITCHES AND AGE CLOCKS: The jaw-drop science of age and disease reversal. Research that makes lifespan just a number

DAVID SINCLAIR *Miracle Man, Geneticist, Harvard Medical School*

DEFYING AGEING



In our lifetimes, we are truly going to see the ability for our doctors to give us medicines that reverse our age and we can do that multiple times

When we're just born, we stay young because there are three genes that are turned on. They're called O, S and K for short. And when we turn on these three genes in the adult tissue, in human adult tissue, or in a mouse, those three genes make the body younger again. This is the reset of the software

At one time, the idea that ageing could be slowed down, let alone reversed, might have sounded crazy, if not impossible. But today, thanks to the work being done at Dr David Sinclair's epigenetics lab—The Sinclair Lab at Harvard Medical School—and other such labs, there is a real possibility of reinstating youthfulness in our tissues and organs.

"What we've discovered is that how we live our life, what we eat, how much we move, the type of molecules that we ingest every day, how we sleep, how relaxed we are, how many friends we have, this modifies another type of system in the body that's called the epigenome," says Sinclair. While the genome is the DNA we inherit from our parents, the epigenome is what controls the DNA and gives instructions to each cell in the body—for example, telling a cell to be a brain cell, or an immune cell, or a toenail and so on. "Over time, we lose those instructions. The epigenetic instructions become lost. And what we've found is that by living a healthy life... you can slow that rate of ageing and prevent this," says Sinclair, "this corruption of the body's software and even reboot it." ■

KEY TAKEAWAYS

- Skipping meals and reducing them down to one or two can actually help turn on genes in your body that fight against ageing
- Exercise, calming your mind down and eating a plant-based diet also help slow down ageing
- Giving our bodies a shock and some stress can actually help reduce the pace at which we age